

VOJENS AP, Denmark

WMO: 061100

Lat: 55.2253N

Lon: 9.2633E

Elev: 141

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	19.8	12.1	10.3	18.2	16.0	12.4	22.6	35.4	44.9	31.8	44.9	4.5	10	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	80.0	66.1	76.5	65.2	73.1	63.3	68.2	75.8	66.4	73.6	64.6	71.1	10.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.9	71.6	63.8	89.1	69.5	62.2	84.2	67.7	32.5	75.6	31.0	73.5	29.7	71.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.3	21.5	DB	8.3	84.4	6.4	3.1	3.7	86.6	-0.1	88.4	-3.7	90.2	-8.3	92.4	(4)
(5)			WB	7.7	71.6	6.2	1.7	3.3	72.9	-0.3	73.8	-3.8	74.8	-8.3	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	33.9	34.4	38.1	45.4	52.1	57.1	61.7	61.7	56.2	48.8	41.2	35.9
	DBStd	11.72	7.01	6.92	6.07	6.06	5.78	5.07	5.19	4.91	4.65	5.66	6.18	7.64
	HDD50	2314	498	437	370	164	44	3	0	0	5	90	267	436
	HDD65	6530	963	857	834	589	401	242	132	128	265	504	714	901
	CDD50	1328	0	0	1	25	109	217	364	364	192	52	4	0
	CDD65	67	0	0	0	0	1	6	31	27	2	0	0	0
	CDH74	558	0	0	0	1	12	67	276	195	7	0	0	0
	CDH80	81	0	0	0	0	0	7	44	30	0	0	0	0
Wind	WSAvg	10.2	12.0	12.0	11.7	10.2	9.6	9.8	8.9	8.6	9.1	9.9	10.3	11.0
Precipitation	PrecAvg	34.4	3.2	2.4	2.2	1.6	2.1	2.8	3.0	3.9	3.3	3.6	3.0	3.3
	PrecMax	42.4	6.5	5.5	4.9	4.5	3.9	5.6	6.8	7.9	6.7	7.9	6.4	6.6
	PrecMin	23.4	0.2	0.6	0.4	0.1	0.5	0.1	0.6	1.5	0.5	1.6	0.6	1.1
	PrecStd	5.0	1.5	1.0	1.1	0.9	0.9	1.3	1.5	1.7	1.5	1.2	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	49.9	57.6	69.8	75.4	80.6	84.3	83.8	74.9	64.6	57.1	51.8
		MCWB	48.5	46.4	49.2	57.4	62.5	65.7	67.2	66.9	65.4	59.8	55.0	50.1
	2%	DB	46.7	46.8	52.4	64.0	70.9	75.4	80.6	79.1	70.2	60.9	54.0	49.7
		MCWB	45.6	45.4	46.8	53.9	59.7	64.3	66.4	65.8	62.2	57.6	52.2	48.7
	5%	DB	44.9	45.0	48.7	59.7	67.0	71.3	77.0	75.3	66.8	58.8	51.8	47.7
		MCWB	43.9	43.8	44.9	51.4	57.5	61.9	65.6	64.6	60.4	56.1	50.0	46.8
	10%	DB	43.1	43.5	46.5	55.7	63.6	67.7	73.3	71.6	64.3	57.1	49.9	46.0
		MCWB	42.2	42.3	44.0	49.7	55.7	59.9	63.5	63.4	59.1	54.7	48.7	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	47.5	50.8	59.1	65.1	68.9	70.8	70.1	66.3	60.6	55.6	50.6
		MCDB	48.9	48.3	54.6	68.3	73.2	76.1	79.4	77.4	72.3	63.2	56.7	51.4
	2%	WB	45.8	45.6	47.7	55.0	61.4	65.7	68.6	67.7	63.5	58.5	52.8	48.6
		MCDB	46.4	46.4	51.1	61.7	67.8	73.2	77.0	74.8	68.6	60.3	54.0	49.3
	5%	WB	44.0	44.1	45.8	52.4	58.9	63.1	66.5	66.0	61.7	56.6	50.5	46.7
		MCDB	44.7	44.9	47.8	58.1	64.9	69.2	74.2	72.7	65.6	58.3	51.4	47.2
	10%	WB	42.2	42.4	44.1	50.3	56.6	60.8	64.5	64.4	60.1	55.1	48.5	44.9
		MCDB	42.9	43.3	45.9	54.8	62.4	66.2	71.1	70.5	63.3	56.9	49.4	45.6
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.9	12.0	15.7	16.9	16.2	16.9	16.6	14.1	11.6	9.1	8.4
		MCDBR	7.7	9.1	16.1	21.3	22.7	22.0	22.8	21.6	17.7	12.7	8.9	7.7
		MCWBR	7.7	7.7	11.2	12.9	13.6	12.8	11.2	10.6	10.8	9.8	8.0	7.6
	5% WB	MCDBR	7.6	8.4	13.7	18.8	19.7	19.3	20.5	18.7	15.0	11.5	8.5	7.6
		MCWBR	7.7	7.5	10.4	12.3	12.7	12.3	11.4	10.2	9.8	9.5	8.1	7.7
Clear-Sky Solar Irradiance	taub	0.321	0.340	0.365	0.397	0.390	0.385	0.401	0.401	0.379	0.357	0.335	0.309	
	taud	2.254	2.291	2.272	2.232	2.311	2.348	2.321	2.333	2.377	2.383	2.313	2.249	
	Ebn at Noon	187	230	253	260	271	273	266	257	247	222	181	163	
	Edh at Noon	18	25	32	38	38	37	37	35	29	23	17	15	
All-Sky Solar Radiation	RadAvg	158	355	771	1288	1669	1753	1640	1317	919	481	196	116	
	RadStd	25	50	70	149	164	139	178	96	97	57	28	11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-265	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-261	N/A	N/A

Nomenclature: See separate page